

ISE 457 PROJECT REPORT

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Group 5 Members

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Comparing Forecasts with The Actual Numbers

Group 5 Members

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Importing Necessary Libraries

- First installed and imported the libraries needed.
- These libraries are necessary for time series analysis, data processing, and visualization.

```
In [ ]: install.packages("fpp3")
install.packages("urca")

Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

also installing the dependencies 'numDeriv', 'ggplot2', 'quadprog', 'warp', 'BH', 'Rcpp', 'distributional', 'progressr', 'ggdist', 'slider', 'anytime', 'fable', 'fabletools', 'feasts', 'tsibble', 'tsibbledata'

Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

In [ ]: library(tidyverse)
library(tsibble)
library(urca)
library(fpp3)
```

Importing Data

- Imported the data files and selected a specific time series according to the student ID which is 20180703056.
- Used the code on YUlearn which converts the data to a timeseries by using tsibble.

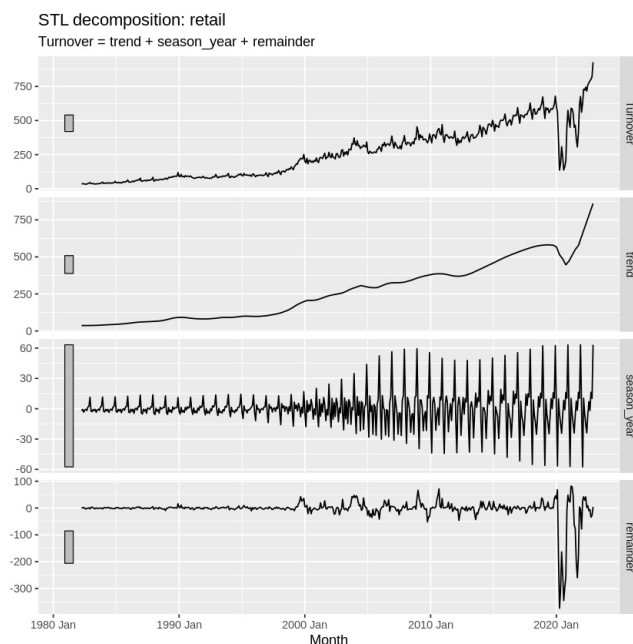
- The section corresponding to "Industry" was "Cafes, restaurants, and catering services."
- Built tests and models based on data.

```
In [ ]: get_my_data2 <- function(student_id) {
  set.seed(student_id)
  all_data <- read.csv("/content/all_data_upto202212.csv")
  data_2023to2024 <- read.csv("/content/data_202301_202402.csv")
  all_data <- all_data |> mutate(Month = yearmonth(Month)) |> as_tsibble(index = Month, key
= c(State, Industry, Series.ID))
  data_2023to2024 <- data_2023to2024 |> mutate(Month = yearmonth(Month)) |> as_tsibble(index = Month, key = c(S
while(TRUE) {
  retail <- filter(all_data, Series.ID == sample(Series.ID, 1))
  if(!any(is.na(fill_gaps(retail)$Turnover)))
  retail2023to2024 <- filter(data_2023to2024, Series.ID == sample(Series.ID, 1))
  if(!any(is.na(fill_gaps(retail2023to2024)$Turnover)))
  return(list(retail, retail2023to2024))
}
}
# Replace the 12345678 argument with your student ID
retail <- get_my_data2(80703056)[[1]]
retail2023to2024 <- get_my_data2(80703056)[[2]]
```

A discussion of the statistical features of the original data, including the effect of COVID-19

STL Decomposition

```
In [ ]: retail %>%
model(STL(Turnover ~ season(window=9), robust=TRUE)) %>%
components() %>% autoplot() +
labs(title = "STL decomposition: retail")
```



This code performs an STL (Seasonal and Trend decomposition using Loess) decomposition on the retail dataset.

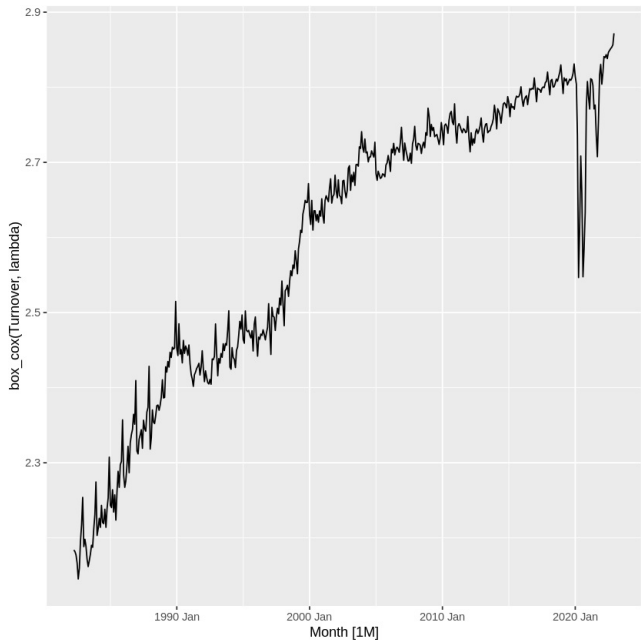
- By applying STL decomposition data set was decomposed.
- There is a general upward trend in the data, with periodic fluctuations and a significant increase around the beginning of 2020.
- The seasonal component contains repeating patterns that occur at specific periods. Over time, these patterns become more pronounced. Yearly changes are increasingly noticeable, especially after the 2000s.
- The part left after removing the trend and seasonal components generally consists of small fluctuations and some sudden changes. A significant anomaly is observed around the beginning of 2020.

Box-Cox Transformation

```
In [ ]: lambda <- retail |>
  features(Turnover, features = guerrero) |>
  pull(lambda_guerrero)
lambda
```

- The Guerrero method was used to determine the lambda parameter that best stabilizes the data for applying the Box-Cox transformation.

```
In [ ]: retail |>
  autoplot(box_cox(Turnover, lambda))
```



- The Box-Cox transformation aims to stabilize the variance of the data. The upward trend is still evident in the transformed data, indicating that the overall increase in Turnover over time is maintained after the transformation.
- The significant drop around 2020 is still present, reflecting the impact of the COVID-19 pandemic. The transformation does not remove such anomalies but stabilizes the data around them.

Unit-Root Test (KPSS Test)

```
In [ ]: retail %>%
  features(Turnover, unitroot_kpss)
```

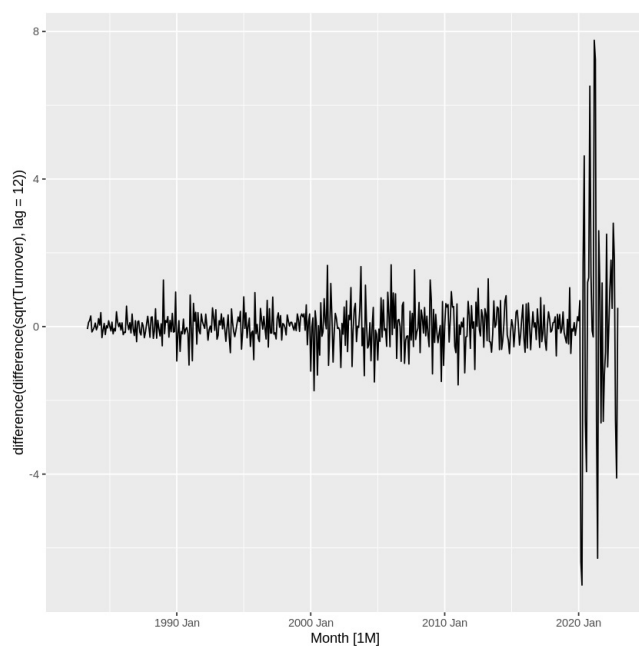
A tibble: 1 × 5

State	Industry	Series.ID	kpss_stat	kpss_pvalue
<chr>	<chr>	<chr>	<dbl>	<dbl>
V	Cafes, restaurants and catering services	A3349640L	7.550697	0.01

- The KPSS test was applied to check if the series is stationary.
- Null Hypothesis H0: The series is stationary.
Alternative Hypothesis H1: The series is non-stationary.
- The test statistic of 7.550697 indicates that there is strong evidence against the null hypothesis that the Turnover data is stationary.
- The p-value is 0.01. If p-value < 0.05, reject the null hypothesis, indicating the series is stationary. According to these results the data is non-stationary.
- As the series is non-stationary, differencing was applied to try to achieve stationarity.

```
In [ ]: retail |>
  autoplot(sqrt(Turnover) |> difference(lag=12) |> difference())
```

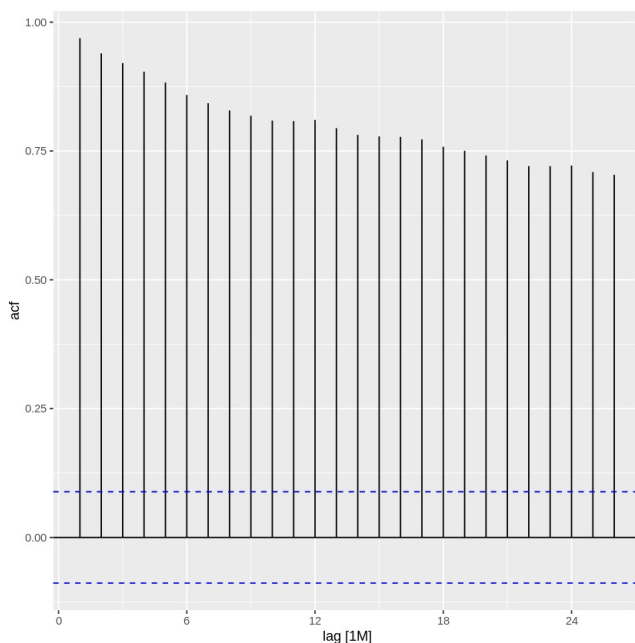
Warning message:
"Removed 13 rows containing missing values or values outside the scale range ('geom_line()')."



- The series appears more stable, with constant variance, although there's increased variability around 2020. This preprocessing is crucial for making the data more suitable for statistical analysis and forecasting.
- This seems to take away the effects of seasonality and the series is looking more like white noise.

Residuals

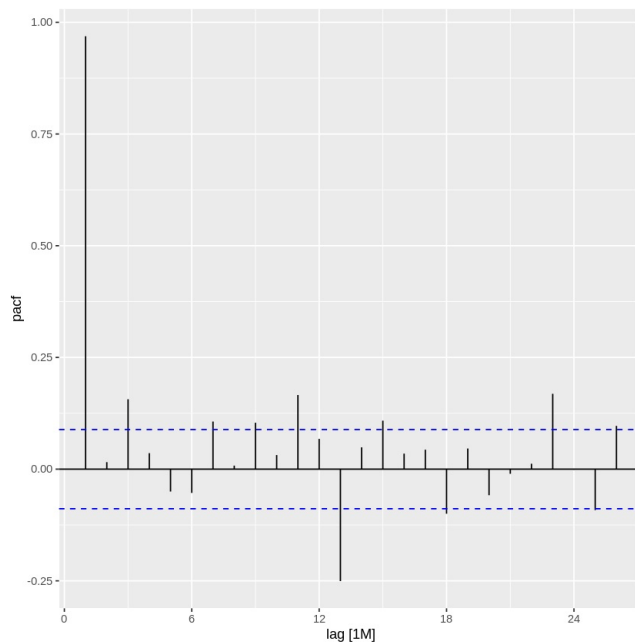
```
In [ ]: retail |>
  filter(Industry == "Cafes, restaurants and catering services") |>
  ACF(Turnover) |>
  autoplot()
```



- The high autocorrelation values and their slow decay indicate that the Turnover data is not stationary and is dependent over time. This suggests a strong autocorrelation structure in the data.
- The autocorrelations outside the blue dashed lines show that these dependencies are statistically significant.

```
In [ ]: retail |>
  filter(Industry == "Cafes, restaurants and catering services") |>
  PACF(Turnover) |>
```

```
autoplot()
```



- High partial autocorrelation values indicate that the data is not stationary and is dependent over time.
- The partial autocorrelations outside the blue dashed lines show that these dependencies are statistically significant.

ETS Model

```
In [ ]: Cafes <- retail %>%  
  filter(Industry == "Cafes, restaurants and catering services")  
  
fit <- Cafes %>%  
  model(ANN = ETS(Turnover ~ error("A") + trend("N") + season("N")))  
  
report(fit)
```

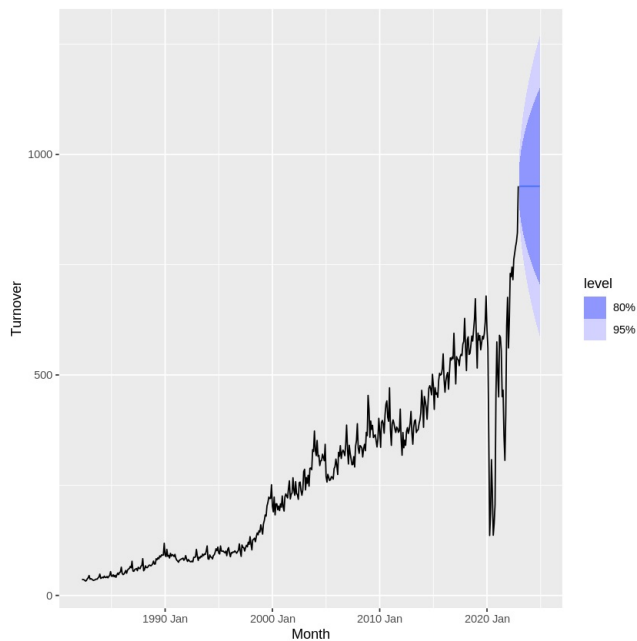
```
Series: Turnover  
Model: ETS(A,N,N)  
Smoothing parameters:  
  alpha = 0.9998997  
  
Initial states:  
  l[0]  
35.84509  
  
sigma^2: 1255.024  
  
      AIC      AICc      BIC  
6521.032 6521.082 6533.609
```

```
In [ ]: fit |> tidy()
```

A tibble: 2 × 6

State	Industry	Series.ID	.model	term	estimate
<chr>	<chr>	<chr>	<chr>	<chr>	<dbl>
V Cafes, restaurants and catering services	A3349640L	ANN	alpha	0.9998997	
V Cafes, restaurants and catering services	A3349640L	ANN	l[0]	35.8450916	

```
In [ ]: fit |> forecast(h = "2 years") |>  
  autoplot(retail)
```



- The Turnover data is expected to continue increasing until 2022. The predicted values indicate that the current upward trend will persist.
- The prediction intervals show that the forecasts are not certain and that future values could vary within the predicted ranges. The uncertainty intervals widen over time, indicating that future forecasts become more uncertain.

ARIMA Model

```
In [ ]: fit <- retail |>
  model(ARIMA(box_cox(Turnover, lambda)))
  report(fit)
```

```
Series: Turnover
Model: ARIMA(0,1,2)(2,0,0)[12]
Transformation: box_cox(Turnover, lambda)

Coefficients:
      ma1      ma2      sar1      sar2
    -0.0675 -0.2448  0.3789  0.2377
s.e.   0.0439  0.0415  0.0441  0.0453

sigma^2 estimated as 0.0004058: log likelihood=1212.63
AIC=-2415.27 AICc=-2415.14 BIC=-2394.31
```

- A suitable ARIMA model was fitted to the transformed data using the ARIMA() function.

```
In [ ]: fit <- retail |>
  model(
    arima010 = ARIMA(box_cox(Turnover, lambda) ~ 1 + pdq(0, 1, 0)),
    arima011 = ARIMA(box_cox(Turnover, lambda) ~ 1 + pdq(0, 1, 1)),
    arima013 = ARIMA(box_cox(Turnover, lambda) ~ 1 + pdq(0, 1, 3)),
    arima110 = ARIMA(box_cox(Turnover, lambda) ~ 1 + pdq(1, 1, 0)),
    arima111 = ARIMA(box_cox(Turnover, lambda) ~ 1 + pdq(1, 1, 1)),
    arima112 = ARIMA(box_cox(Turnover, lambda) ~ 1 + pdq(1, 1, 2)),
    arima113 = ARIMA(box_cox(Turnover, lambda) ~ 1 + pdq(1, 1, 3)),
    arima210 = ARIMA(box_cox(Turnover, lambda) ~ 1 + pdq(2, 1, 0)),
    arima211 = ARIMA(box_cox(Turnover, lambda) ~ 1 + pdq(2, 1, 1)),
    arima212 = ARIMA(box_cox(Turnover, lambda) ~ 1 + pdq(2, 1, 2)),
    arima310 = ARIMA(box_cox(Turnover, lambda) ~ 1 + pdq(3, 1, 0)),
    arima311 = ARIMA(box_cox(Turnover, lambda) ~ 1 + pdq(3, 1, 1))
  )
```

- Other plausible models were also tried.

```
In [ ]: fit |>
  glance() |>
  arrange(AICc) |>
  select(.model, AICc)
```

A tibble: 12 × 2

.model	AICc
<chr>	<dbl>
arima212	-2404.171
arima010	-2381.919
arima310	-2381.341
arima311	-2379.491
arima211	-2378.818
arima210	-2376.835
arima013	-2372.338
arima112	-2372.241
arima113	-2371.784
arima111	-2366.523
arima110	-2352.104
arima011	-2343.031

- According to the AIC values of the models, it was determined that ARIMA(0,1,2)(2,0,0) is the best fit for the given data because it has the smallest AIC.

```
In [ ]: best_fit <- retail |>
         model(ARIMA(box_cox(Turnover, lambda) ~ 1 + pdq(0, 1, 2)))
         best_fit |> report()
```

Series: Turnover

Model: ARIMA(0,1,2)(0,0,2)[12] w/ drift

Transformation: box_cox(Turnover, lambda)

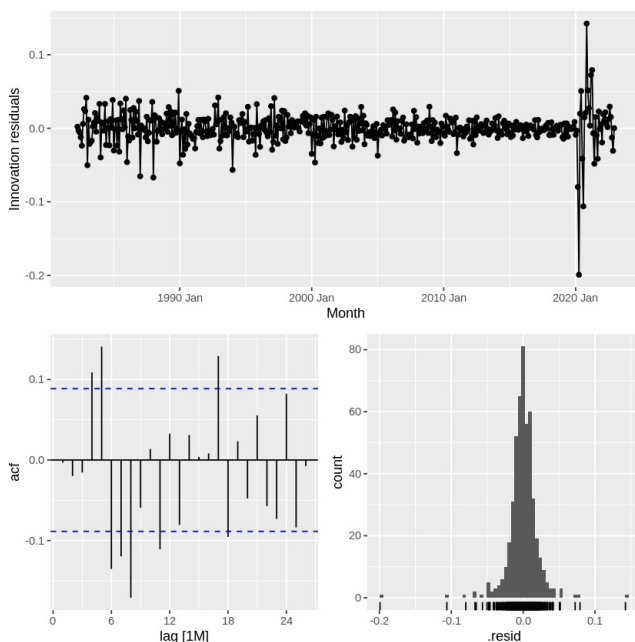
Coefficients:

	ma1	ma2	sma1	sma2	constant
	-0.1229	-0.2508	0.376	0.2364	0.0015
s.e.	0.0446	0.0442	0.046	0.0442	0.0009

sigma^2 estimated as 0.0004424: log likelihood=1193.2

AIC=-2374.4 AICc=-2374.22 BIC=-2349.25

```
In [ ]: best_fit |> gg_tsresiduals()
```



- There are some large deviations around 2020, likely due to the impact of the COVID-19 pandemic.
- Some autocorrelations are outside the blue lines, suggesting that the model may not capture some temporal dependencies adequately.
- Most residuals are clustered around zero, indicating good overall model performance, but there are some outliers, suggesting that the model doesn't fully capture some observations.

Ljung-Box Test

```
In [ ]: augment(best_fit) |> features(.innov, ljung_box, dof = 1, lag = 10)
```

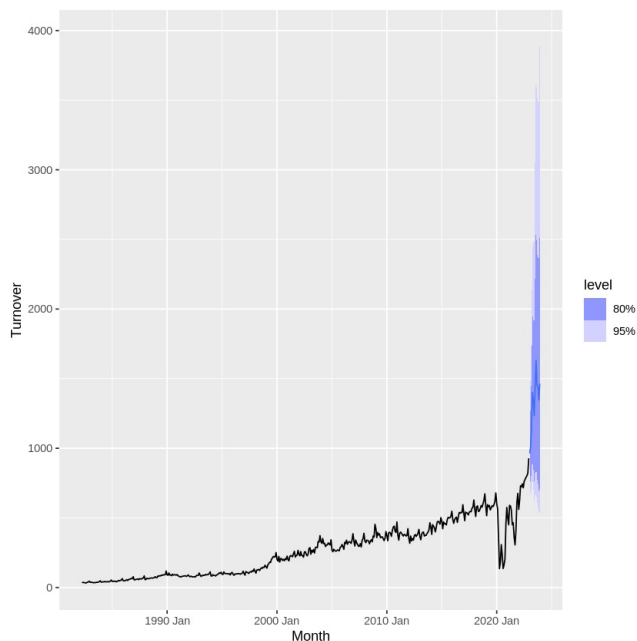
A tibble: 1 × 6

State	Industry	Series.ID	.model	lb_stat	lb_pvalue
<chr>	<chr>	<chr>	<chr>	<dbl>	<dbl>
V	Cafes, restaurants and catering services	A3349640L	ARIMA(box_cox(Turnover, lambda) ~ 1 + pdq(0, 1, 2))	48.71769	1.874822e-07

- The Ljung-Box test statistic value is high, which is 48.71769.
- The p-value associated with the Ljung-Box test statistic is so low, which is 1.874822e-07.
- So there is significant autocorrelation in the residuals of the best_fit model. This indicates that the residuals are not completely random and that the model may need improvement to better capture the dependencies in the data.

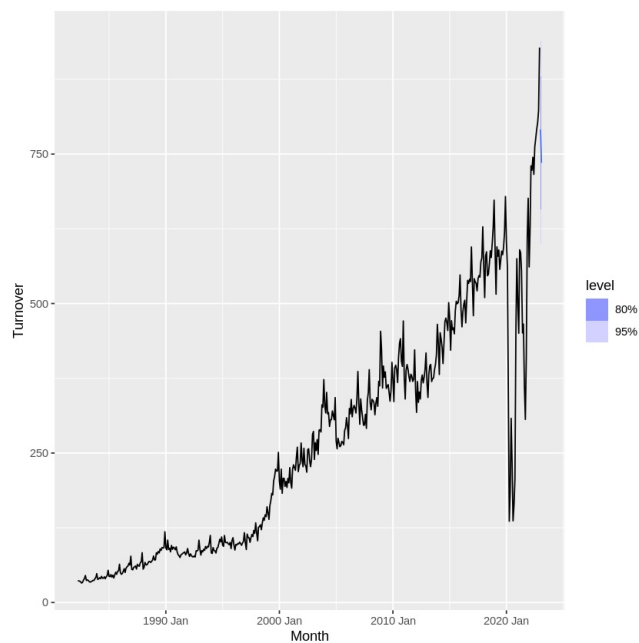
Forecast

```
In [ ]: best_fit |>
  forecast(h = 12) |>
  autoplot(retail)
```



- The Turnover data is expected to continue increasing significantly over the next 12 months. The predicted values indicate that the current upward trend will persist.
- The prediction intervals show that the forecasts are not certain and that future values could vary within these ranges. The widening of the uncertainty intervals over time indicates that longer-term forecasts are less certain and contain more uncertainty.

```
In [ ]: retail |>
  model(ETS(Turnover)) |>
  forecast(h = 2) |>
  autoplot(retail)
```



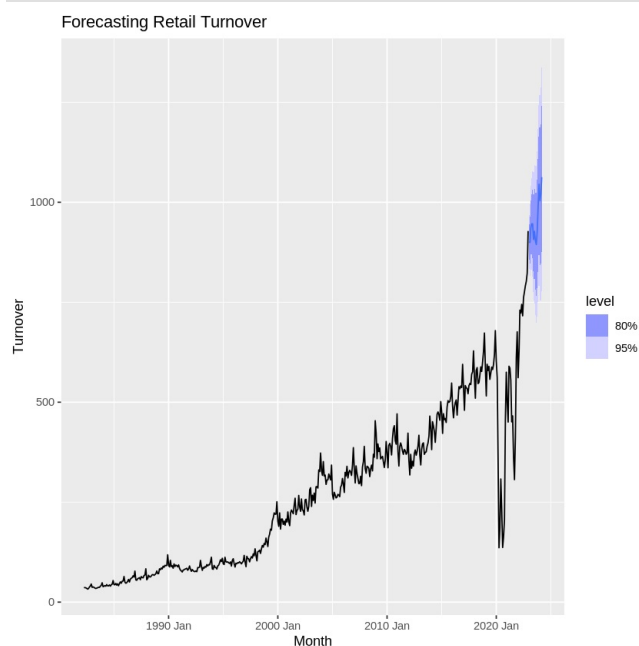
- This graph provides a 2-year forecast for the Turnover data and shows the prediction intervals. The forecasted values indicate that the current upward trend will continue, with a certain level of uncertainty.
- The prediction intervals show the possible range of future values, with the uncertainty increasing over time. Such forecasts can help in making business decisions and future planning.

Comparing Forecasts with The Actual Numbers

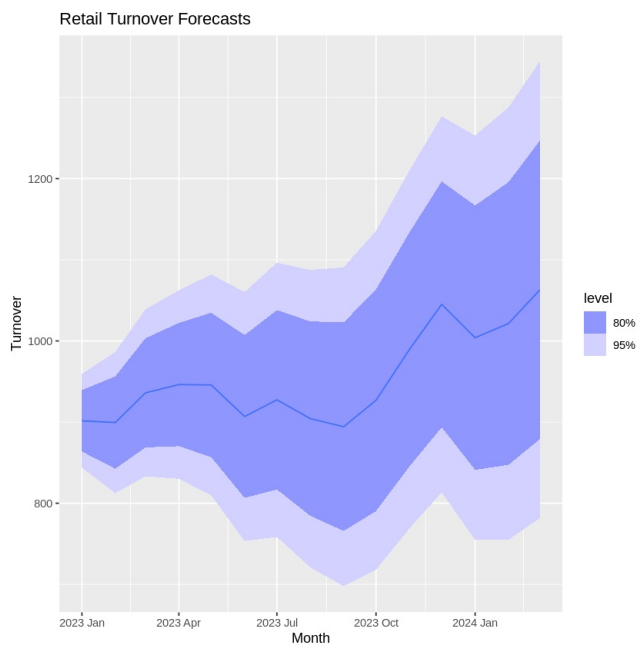
```
In [ ]: best_fit <- retail %>%
  model(
    arima = ARIMA(Turnover)

fc <- best_fit %>%
  forecast(h = "15 month")

fc %>%
  autoplot(retail) +
  labs(title = "Forecasting Retail Turnover",
    y = "Turnover",
    x = "Month")
```



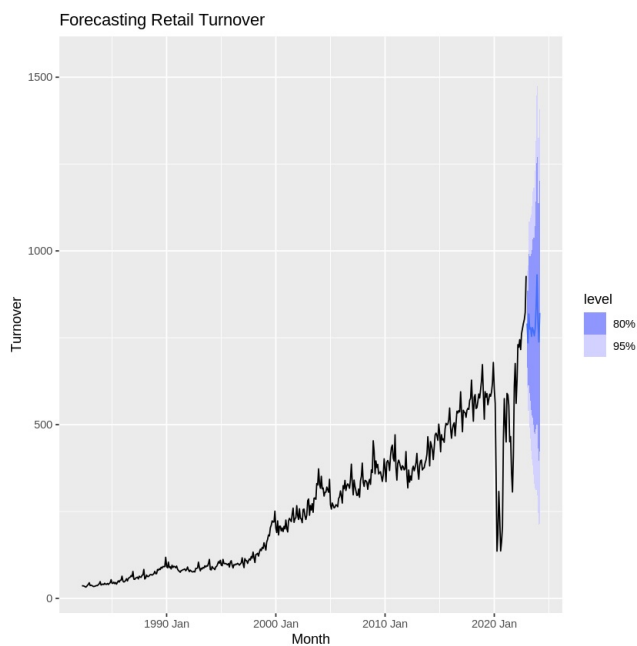
```
In [ ]: autoplot(fc) +
  labs(title = "Retail Turnover Forecasts",
    y = "Turnover",
    x = "Month")
```



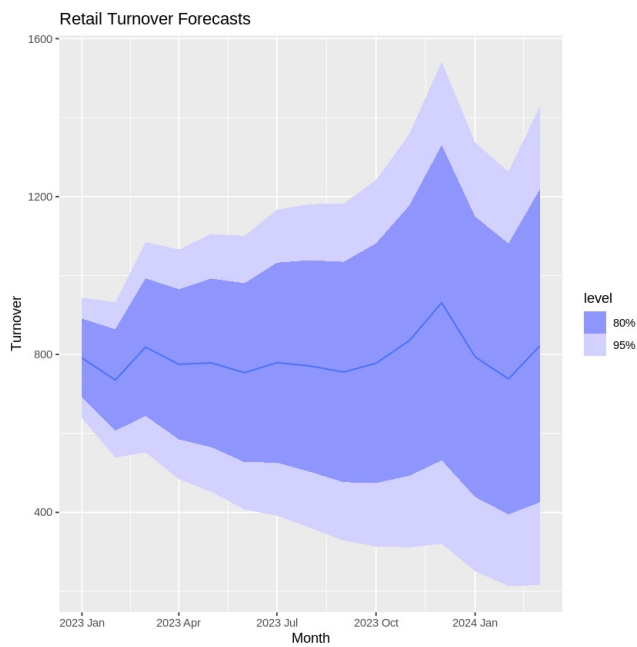
```
In [ ]: fit <- retail %>%
  model(
    ets = ETS(Turnover)
  )

fce <- fit %>%
  forecast(h = "15 month")

fce %>%
  autoplot(retail) +
  labs(title = "Forecasting Retail Turnover",
    y = "Turnover",
    x = "Month")
```

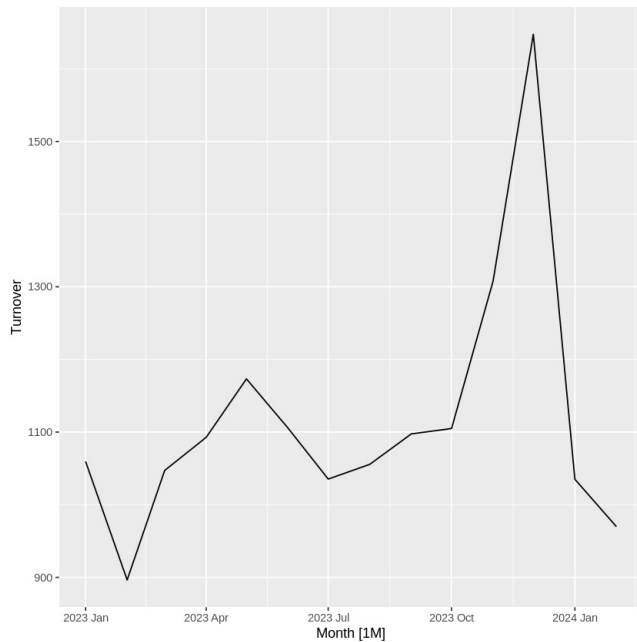


```
In [ ]: autoplot(fce) +
  labs(title = "Retail Turnover Forecasts",
    y = "Turnover",
    x = "Month")
```



```
In [ ]: autoplot(etail2023to2024)
```

Plot variable not specified, automatically selected `.vars = Turnover``



- As seen from the graphs, the predictions made with the ETS model yielded results more similar to the real data.
- The Turnover values are expected to generally increase throughout 2023 and into early 2024.
- Prediction Confidence Intervals: The forecasted values fall within certain intervals, and these intervals widen over time, indicating increasing uncertainty.
- The forecasts are based on existing data trends, and the model attempts to predict future trends.

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